

Polynomially many ordered rows with a fixed number of rational rank columns

Research proof: independently reviewed

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Status. The full proof, including the continuous-density baseline, has been independently reviewed by the root and rational-design research agents.

Scope. This is an auxiliary moment theorem for a fixed number of rational cumulative-rank columns. It does not establish the ordered-prefix identities or positive joint completion needed for three or more simultaneous small dice.

Lemma 1 (A lacunary fully fair continuous baseline). *Fix $u \geq 0$ and $k \geq 1$. Suppose positive integers d_i satisfy*

$$d_i > u + k - 1 + \sum_{j < i} d_j.$$

For $|\eta_i| < 1$, let the i th exceptional continuous variable have density $f_i(t) = 1 - \eta_i P_{d_i}(2t - 1)$ on $[0, 1]$. Together with u independent uniform variables, these k independent variables are fully permutation-fair. All densities are strictly positive; if the η_i are rational, their coefficients are rational.

Proof. Fix one complete order and expand the product of the k densities. The empty correction term is its all-uniform probability $1/(u+k)!$. For a nonempty correction subset, choose its largest index i and condition on that variable having value t . Integrating all other ordered variables below and above t gives an ordinary polynomial in t of degree at most

$$u + k - 1 + \sum_{j < i} d_j < d_i.$$

Here each integration adds at most one to the total degree, and only the selected lower-index perturbations contribute their degrees. Legendre orthogonality makes the remaining integral against $P_{d_i}(2t - 1)$ zero. Thus every nonempty term vanishes. Positivity and normalization follow from $|P_{d_i}| \leq 1$ and its zero mean. $\square$

Theorem 1. *Fix integers $k \geq 1$ and $T \geq 1$. For every $m \geq 1$, there exist $K = O_k((m+1)^4)$ ordered real nodes $0 < b_1 < \dots < b_K < 1$ and k strictly increasing rational columns $0 < r_{i,1} < \dots < r_{i,K} < 1$ such that*

$$\frac{1}{K} \sum_{q=1}^K b_q^a \prod_{i \in S} r_{i,q} = \frac{1}{a + |S| + 1}, \quad S \subseteq [k], \quad 0 \leq a \leq m. \quad (1)$$

All ranks have one common denominator at most $C_{k,T}(m+1)^{C_{k,T}}$. The nodes have consecutive and endpoint gaps at least c_k/K , and

$$\max_{i,q} |r_{i,q} - b_q| \leq C_{k,T}(m+1)^{-T}.$$

All constants are independent of m .

The proof uses two independently reviewed inputs: the separated midpoint-grid quadrature lemma in `../rational_designs/asymptotic/two_discrete_step_compatibility.tex` and the polynomial conditioning theorem in `../rational_designs/asymptotic/lacunary_legendre_conditioning.tex`. For fixed k, T , below “polynomial” means bounded by a fixed power of $m + 1$, with constants and exponent allowed to depend on k, T .

Proof. Put $H_d(t) = \int_0^t P_d(2u - 1) du$. It vanishes at both endpoints, has derivative bounded by one, and is orthogonal in Lebesgue measure to every polynomial of degree at most $d - 2$. For example the latter fact follows from its expression as a linear combination of shifted Legendre polynomials of degrees $d - 1$ and $d + 1$. Choose integers

$$d_i = m + k + 2 + \sum_{j < i} (d_j + 1), \quad 1 \leq i \leq k,$$

so all degrees are $O_k(m + 1)$, and put

$$R_i(t) = t - \eta H_{d_i}(t), \quad \eta = c(m + 1)^{-(T+1)}$$

for a fixed sufficiently small positive rational c . These maps increase from zero to one with derivative at least $1/2$. Expanding a product and integrating the factor with largest index in each nonempty correction term gives

$$\int_0^1 t^a \prod_{i \in S} R_i(t) dt = \frac{1}{a + |S| + 1}, \quad a \leq m. \quad (2)$$

Indeed the polynomial multiplying the largest H_{d_i} has degree at most $m + k + \sum_{j < i} (d_j + 1) \leq d_i - 2$.

Let $D = m + \sum_i (d_i + 1) = O_k(m + 1)$. Choose the reviewed real equal-weight quadrature exact through degree $2D$, with $K = O_k((m + 1)^4)$ nodes b_q^0 at distance at most $1/(8K)$ from the uniform midpoint grid. Set $r_{i,q}^0 = R_i(b_q^0)$. All identities (1) hold at this real baseline, and all rank and node gaps are at least c_k/K . In fact this real baseline has a stronger full-order interpretation: take the k densities in Lemma 1 with these degrees and parameters, together with $m + 1$ uniforms, and replace one uniform by the quadrature variable B . Its conditional complete-order tests have degree at most $m + k + \sum_i d_i = D$, so the quadrature preserves all complete orders. The subsequent rational-rank construction below retains the stated mixed threshold moments; no claim that it retains this stronger full-order interpretation is made.

Rationally fixing the constant mixed moments. Choose an integer L_0 of sufficiently large polynomial size and round all $r_{i,q}^0$ to multiples of $1/L_0$. This makes an error at most $1/L_0$ in each entry. We next correct exactly the finitely many identities (1) with $a = 0$ and $S \neq \emptyset$.

Partition the row indices into k interlaced groups $Q_1, \dots, Q_k$. Proceed with $j = k, k-1, \dots, 1$. At stage j , vary column j only on a fixed number $d_j^* = 2^{k-j}$ of selected rows in Q_j , to impose the equations for all S whose least element is j . These equations are exactly linear in the proposed changes; their coefficient rows are the evaluations of

$$\prod_{i \in U} r_i, \quad U \subseteq \{j + 1, \dots, k\}.$$

Previously fixed equations do not involve column j and are unchanged. Because the row groups are disjoint, all other entries on the rows of Q_j still equal their initially rounded values.

We justify a polynomially conditioned square minor. At the baseline, the corresponding continuous polynomial family is independent and has polynomial least singular value by the reviewed lacunary rank-product theorem. Sampling on Q_j preserves this up to a constant: its star discrepancy is $O_k(1/K)$, and for any polynomial F of degree at most D ,

$$\left| \frac{1}{|Q_j|} \sum_{q \in Q_j} F(b_q^0)^2 - \int F^2 \right| \leq CD^2 \text{disc}(Q_j) \|F\|_2^2.$$

The right side is at most half the squared norm after the absolute constant in K is increased. Initial rounding preserves polynomial coercivity if L_0 is a sufficiently large power. The Cauchy–Binet formula now supplies a $d_j^* \times d_j^*$ evaluation minor with polynomially bounded inverse: its dimension depends only on k , and its entries are bounded. Solving that rational minor corrects all equations at this stage with changes of size at most $\text{poly}(m)/L_0$.

The denominator increase is also polynomial. Each matrix entry has denominator dividing a fixed power of L_0 , its dimension is fixed, and its determinant numerator is bounded by a fixed power of L_0 . At successive stages, the right sides involve only the finitely many already corrected entries and the targets $1/(|S|+1)$. Cramer’s rule therefore leaves a polynomial common denominator after every stage. Only $\sum_j 2^{k-j} = 2^k - 1$ entries have been altered beyond initial rounding. Denote the resulting rational array by r ; it obeys all constant mixed moments exactly and differs from r^0 by $\text{poly}(m)/L_0$.

Correcting all nonconstant moments by moving the nodes. Let $\phi_0, \dots, \phi_{m-1}$ be the orthonormal shifted Legendre polynomials, and let $f_a(t) = \int_0^t \phi_{a-1}(u) du$ for $1 \leq a \leq m$. With r fixed, impose

$$\frac{1}{K} \sum_q f_a(b_q) \prod_{i \in S} r_{i,q} = \int_0^1 f_a(t) t^{|S|} dt, \quad S \subseteq [k], \quad 1 \leq a \leq m. \quad (3)$$

Together with the corrected constant equations, these are equivalent to (1), because $1, f_1, \dots, f_m$ span all degree- m polynomials.

At (b^0, r^0) the Jacobian in b , multiplied by K , evaluates

$$\phi_{a-1}(t) \prod_{i \in S} R_i(t), \quad 1 \leq a \leq m, S \subseteq [k].$$

Its exact empirical Gram is the continuous Gram, since the baseline quadrature integrates all products through degree $2D$. The lacunary conditioning theorem gives a polynomial lower eigenvalue. The number of rows, all evaluation norms, and their first derivatives are polynomially bounded. Consequently the fixed right inverse of this baseline Jacobian has polynomial norm from Euclidean residual norm to maximum node displacement. The rank rounding gives residual $\text{poly}(m)/L_0$; the change of the Jacobian on a ball of maximum radius ρ is at most $\text{poly}(m)(\rho + L_0^{-1} \text{poly}(m))$.

The same fixed-inverse contraction argument as in the reviewed two-column construction therefore solves (3) within maximum displacement $\text{poly}(m)/L_0$, provided L_0 is chosen to be a sufficiently large fixed power of $m + 1$. For clarity, if A is the baseline Jacobian and B a fixed right inverse, use $b \mapsto b - BF(b, r)$ on $b^0 + \text{range}(B)$. There BA is the identity; the preceding derivative estimate makes this map a contraction preserving a ball of twice its initial displacement. No arithmetic condition is needed on these corrected nodes.

Increase the exponent of L_0 so that both rank and node changes are less than $c_k/(10K)$ and also $o((m+1)^{-T})$. All ranks and nodes then remain strictly ordered and interior, with the asserted gaps. Since $|R_i(t) - t| \leq \eta$, their mutual closeness has the required scale. The rational common denominator remains polynomial by the constant-dimensional correction argument above. This proves the theorem. $\square$

What this does not encode. The products in (1) specify simultaneous threshold events, not the internal order of the k variables below or above a threshold. For $k \geq 2$, a simultaneous small-dice construction additionally needs ordered prefix integrals and a positive completion preserving them. No such implication is asserted here. The theorem isolates a genuine arithmetic and conditioning subproblem from that remaining order issue.